Structured residual learning for combinatorial perturbation prediction: an internal benchmark of Opal v1

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Abstract

Predicting the mean transcriptomic response to an unmeasured combination is a useful, restricted task when both component perturbations have been measured in the same context. Opal represents the combined effect as the sum of observed single effects plus a symmetric learned correction. We implement coordinate-product, learned bilinear, and symmetric neural corrections, with restricted and unrestricted output representations, and compare them against common-information classical controls. Eight synthetic scenarios test recovery, scale dependence, shared distortions, emergent effects, and measurement noise. The selected models recover structured synthetic residuals, but a global correction outperforms the neural selection when the generator contains only a global distortion. On Norman CRISPR activation data, the selected rank-two symmetric network reduces held-out mean squared error by 15.81% relative to additive prediction, with an inconclusive advantage over its development-selected classical comparator. On original-source THP1 Cas13 data, the selected rank-eight network without multiplicative features reduces error by 61.08%, while a development-selected global correction reduces error by 69.76% and outperforms it. These results do not establish a biological advantage from pair-product features. Original ComboSciPlex supports only seven combinations with both singles, preventing the intended independent chemical evaluation. Exploratory nested development amendments favor a residual kernel over the tested neural additions. On a fixed original Wessels validation population, adding eligible training pairs reduces random-fold error by 16.05% and partner-fold error by 4.92%; these are coverage gains with known response histories, not independent confirmation. The present evidence illustrates why improvement over additivity alone is insufficient to establish a structural advance. Public-method comparisons and replication-aware uncertainty remain necessary for the full v1 research claim.

Keywords: combinatorial perturbations; transcriptomic prediction; residual learning; symmetric models; reproducibility; evaluation.

1 Introduction

Single-cell perturbation experiments connect interventions to transcriptional responses, while combinatorial screens measure departures from single-intervention behavior. Norman's K562 CRISPR activation experiment provides a widely used genetic combination dataset (Norman et al., 2019). CaRPool-seq extends combinatorial targeting to RNA transcripts using Cas13 and provides a second experimental system (Wessels et al., 2023). These experiments motivate a prediction task that is narrower than general perturbation prediction: given measured single profiles in one context, predict an unmeasured simultaneous pair.

The additive prediction supplies a strong starting point. A learned model can improve on it by capturing a shared calibration error, a common offset, or pair-specific structure. Only the last explanation supports a claim that particular pair features contribute useful information. Metric calibration also matters because errors across many weakly responding genes and correlations dominated by single effects can obscure the quality of interaction prediction (Miller et al., 2026). We therefore compare the proposed correction to stronger controls using the same observed singles, and assess synthetic negative controls before interpreting biological gains.

This work implements Opal v1 as a family of output-level residual models and an auditable experimental pipeline. The present contribution is the controlled internal comparison and its negative biological finding, rather than a demonstrated state-of-the-art method. Residual subtraction, symmetry, and dimensionality restriction are established modeling tools. The experiments ask which of these restrictions help under a frozen selection procedure, and retain unfavorable results. The complete research scope also requires independent genetic confirmation, chemical validation, public-method comparisons, and fuller uncertainty assessment; the current manuscript records their incomplete status.

2 Related work

CPA learns compositional representations for predicting responses to complex perturbations and provides the chemical experiment considered here (Lotfollahi et al., 2023). GEARS uses biological knowledge graphs to predict multigene perturbation responses (Roohani et al., 2024). Neither a published score under a different split nor access to external knowledge is interchangeable with a comparison that supplies observed singles under our protocol. Our internal controls use the same inputs; CPA and GEARS remain unexecuted public-method comparisons.

NAIAD combines single-effect information and learned representations for combinatorial experimental discovery (Qin et al., 2025). Its phenotype and acquisition objectives constrain claims of novelty for any single-informed pair model, but a transcriptomic comparison requires compatible outputs and data access. SLIM supplies a small linear reference using STRING embeddings (Hu et al., 2026); its upstream resource must be accounted for separately. PertAdapt adapts single-cell foundation models to perturbation prediction (Bai et al., 2026), so upstream pretraining and combination-task compatibility must also be disclosed. These references motivate future controlled benchmarks; no result in this manuscript ranks Opal against these methods.

3 Methods

3.1 Task and expression scale

Let $\mu_{u,s,g}$ denote the mean library-normalized count for intervention u, matched stratum s, and gene g, and let $\mu_{0,s,g}$ denote its control mean. Each cell is normalized to the median retained control library total. The primary effect is

$$\delta_{u,g} = \frac{1}{|S_u|} \sum_{s \in S_u} \left[\log_2(1 + \mu_{u,s,g}) - \log_2(1 + \mu_{0,s,g}) \right]. \tag{1}$$

We transform condition means, rather than averaging transformed cell counts. Strata contribute equally, and a retained pair must share stratum support with both component singles. The source-specific stratum definitions are technical matching variables, not assumed independent biological replicates.

For two distinct simultaneous interventions A and B, define $a_{AB} = \delta_A + \delta_B$ and $r_{AB} = \delta_{AB} - a_{AB}$. The prediction is

$$\widehat{\delta}_{AB} = a_{AB} + f_\theta(\delta_A, \delta_B), \qquad f_\theta(\delta_A, \delta_B) = f_\theta(\delta_B, \delta_A). \tag{2}$$

All components have observed single profiles in the same experimental context. The protocol does not evaluate unseen singles, different cell types, different doses, sequential treatments, or individual-cell trajectories. Additivity depends on the chosen expression scale, and r_{AB} also includes error in the estimated singles and controls. It is a prediction residual, not automatically drug synergy or a causal genetic mechanism.

Residual and response losses are identical when the observed additive term is fixed:

$$\|f_\theta - r_{AB}\|_2^2 = \|a_{AB} + f_\theta - \delta_{AB}\|_2^2. \tag{3}$$

Any benefit from residual training therefore arises from parameterization, initialization, or regularization, rather than a different squared-error objective.

3.2 Source preparation and input access

Norman uses the named secondary scPerturb release from Zenodo record 7278143 version 1.1. Its documented transformation reads the original count matrix without expression normalization. We retain source-annotated good-coverage singlet assignments and explicitly negative-control constructs, preserve guide provenance, and match by gemgroup. This is distinct from a GEARS compatibility track. The initial library range is 500–20,000 counts, with at least 50 cells per condition. After source-specific exclusions and target-pair aggregation, the prepared dataset has 105 singles, 126 pairs, and 10,623 eligible genes; 88 pairs are development conditions and 38 are final test conditions.

Wessels uses the original GSE213957 four-lane archive and THP1 metadata (Wessels et al., 2023). Barcode joins retain metadata-matched THP1 Gene Expression measurements and exclude HEK293 cells and guide-count features. Exact `NT` tokens determine controls, avoiding substring misclassification of target names. The minimum library is 500; the maximum is the 99.5th percentile of controls and singles. Conditions require at least 50 cells, and lanes supply matched strata. Many measured singles fail this minimum, so the accepted task covers 16 singles and 58 pairs, with 10,590 genes, 40 development pairs, and 18 test pairs. This restricted coverage is a consequence of the observed-single contract.

Gene eligibility uses controls and singles only. A gene survives if at least one reference condition has normalized mean at least 0.1 and detection fraction at least 0.01. This frozen universe is the primary output; it does not establish performance on genes excluded by the reference rule. Source hashes, count locations, guide mapping, exclusions, normalization parameters, gene order, and split manifests are retained. Held-out pair responses occupy a separate evaluator bundle and are unavailable to model fitting or architecture selection.

3.3 Representations and correction families

PCA fitted to centered single effects supplies an embedding $e_u \in \mathbb{R}^m$. Its dimension is capped by the number of available singles and output genes; nominal dimensions are 32 for Norman and 16 for Wessels and the synthetic suite. An identity representation and a linear denoising encoder are comparison candidates. When split-half single profiles exist, the encoder learns reciprocal reconstruction between halves. Otherwise, the synthetic comparison uses added noise. Joint fine-tuning updates the target embedding table with a single-profile reconstruction penalty; it is not a demonstrated encoder for previously unmeasured targets.

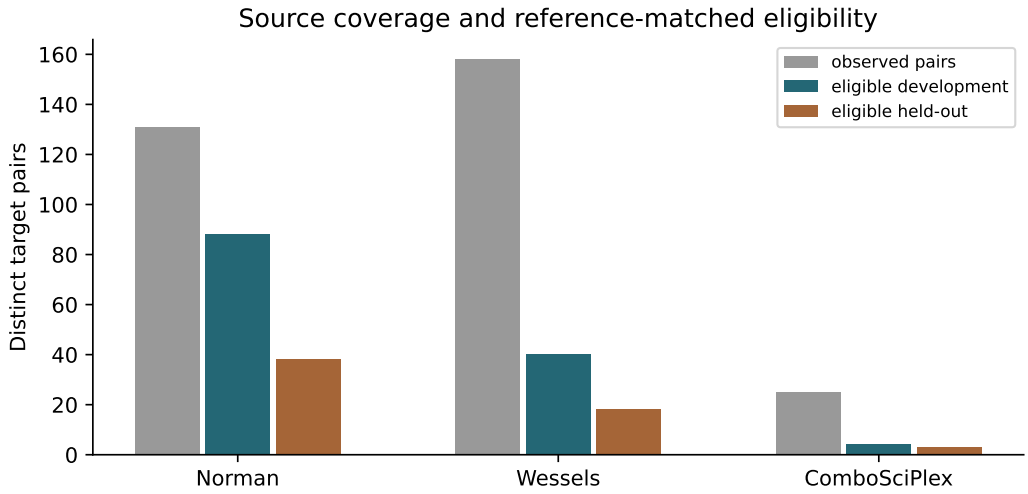


Figure 1: Original distinct nonrepeat pair coverage and surviving reference-matched development/test conditions. Source-level measurement does not guarantee a usable pair with both matched singles.

The coordinate-product model uses $q_{AB} = e_A \odot e_B$. A learned bilinear candidate uses

$$q_{AB} = (Pe_A) \odot (Pe_B), \qquad f_\theta = b + B\Lambda^\top q_{AB}, \tag{4}$$

where $B \in \mathbb{R}^{G \times k}$ defines a shared output space. An independent symmetric-form candidate uses coordinates $e_A^\top W_j e_B$ with symmetric W_j. The neural network candidate applies a tanh network to concatenated $e_A + e_B$, $|e_A - e_B|$, and projected products. Its sum-network control omits the projected-product block but retains the sum and absolute difference. We therefore call this control a network without multiplicative features, rather than claiming that it receives only the sum. Both networks predict coefficients in the output basis plus an optional offset. Unrestricted-output, fixed-basis, zero-offset, random-initialization, direct-response, representation, and fine-tuning comparisons test individual choices.

3.4 Baselines and development selection

The internal baseline set includes no-change, additive, training-mean response, training-mean residual, symmetric ridge regression, reduced-rank regression, nearest-neighbor residual correction, and global correction. Symmetric regression uses sum, absolute difference, and coordinate products of the same embeddings. Global correction uses the observed additive response directly, with an optional feature expansion comprising its coordinates, their hyperbolic tangents, and their squares. It can explain shared nonlinear distortions without using pair identities.

Random pair holdout reserves 30% of canonical unordered pairs, using identities alone. Five development folds determine candidate errors and stopping epochs. The initial CPU search uses ranks 2 and 8, ridge penalties 1 and 100, and at most 80 epochs for neural candidates. AdamW uses learning rate 0.01 and weight decay 0.01, with full-batch equal-condition loss. Initial screening uses one optimization seed; real-data finalist evaluation uses five seeds, while synthetic engineering finalists use two. Final neural predictions average five refitted seeds. The selected neural family and the selected simpler comparator are chosen separately on development error and fixed before

final evaluation. Their comparison can therefore reject the neural structural claim even when the network beats additivity.

Within each development fold, the validation responses choose the best neural stopping epoch and score that checkpoint. Candidate scores average member-specific errors across seeds, rather than scoring a seed-averaged ensemble. Final refits use the median selected epoch across folds and seeds, then average predictions from five seeds before scoring. Development errors are model-selection scores, not unbiased estimates of final ensemble error. For partner cross-validation, predictions from overlapping validation folds are averaged per covered pair before computing each seed's score; uncovered pairs do not contribute.

This bounded search is the initial registered implementation batch, not the roadmap's full capacity expansion. Learning curves use nested development subsets and three subset orders while keeping the full-development-selected hyperparameters. They measure fixed-hyperparameter sensitivity, rather than a subset-retuned sample-efficiency optimum. Additional combination-partner holdouts remove all development pairs incident to specified held targets while preserving their single profiles. Partner folds may overlap in their test pairs and cannot be counted as independent experiments.

3.5 Endpoints and uncertainty

For N test pairs and G frozen genes, the primary endpoint and additive skill are

$$\mathrm{MSE} = \frac{1}{NG}\sum_{p,g}(\widehat{\delta}_{p,g} - \delta_{p,g})^2, \qquad \mathrm{skill}_{\mathrm{add}} = 1 - \frac{\mathrm{MSE}_{\mathrm{model}}}{\mathrm{MSE}_{\mathrm{additive}}}. \tag{5}$$

Pairs have equal weight regardless of cell count. Secondary artifacts record absolute error, response correlation, residual alignment, component-informed panel error, and residual geometry. Component panels use single effects and control dispersion where available, without examining held-out doubles. Evaluator-only emergent-response analyses do not define the fitting gene universe.

Paired improvement is baseline MSE minus selected-model MSE. We report 2,000-draw pair bootstrap and target-product-weight bootstrap sensitivities. These are descriptive intervals conditional on estimated profiles and the fitted selection. They do not propagate shared-control uncertainty, the full measurement process, model selection, or independent biological replication. Split-half profiles provide a measurement-reliability diagnostic, not an accuracy ceiling. Ensemble dispersion is not a calibrated confidence interval or a validated abstention policy.

3.6 Synthetic engineering suite

The suite uses 24 targets, 96 genes, and all 276 unordered distinct pairs. Eight generators produce log-scale additivity, count-scale additivity evaluated on a log scale, low-rank interactions, cross-coordinate interactions, higher-rank interactions, a shared nonlinear additive-response distortion, emergent effects in genes without single effects, and noisy singles with a shared control error. Pair responses receive Gaussian observation noise with standard deviation 0.015. Five total random splits and five partner folds are completed for the low-rank generator. These constructed tasks validate implementation behavior and identifiable failure modes; they do not establish biological performance or prevalence of low-rank interactions.

4 Results

4.1 Synthetic recovery depends on the generating structure

Scenario	Selected family	MSE	Skill	Simpler MSE
additive log	bilinear	0.00022	-0.000	0.00022
additive count	bilinear	0.00050	0.471	0.00054
low rank	network	0.00035	0.968	0.00598
cross coordinate	network	0.00026	0.956	0.00240
high rank	bilinear	0.00943	0.657	0.01134
global distortion	network	0.00110	0.941	0.00104
emergent	network	0.00024	0.859	0.00085
noisy controls	network	0.00241	0.842	0.00562

Table 1: Completed synthetic suite. Scores are equal-pair held-out mean squared errors and additive skill as a fraction. These are engineering tasks, not biological validation.

The log-additive negative control has selected MSE 2.22709×10^{-4} and additive skill -0.019%, so residual learning supplies no useful gain. The low-rank and cross-coordinate generators yield skills of 96.76% and 95.63%, respectively, for development-selected symmetric networks. Their selected errors are 3.48993×10^{-4} and 2.57490×10^{-4}. This demonstrates recovery when the synthetic structure is learnable under the implemented information access.

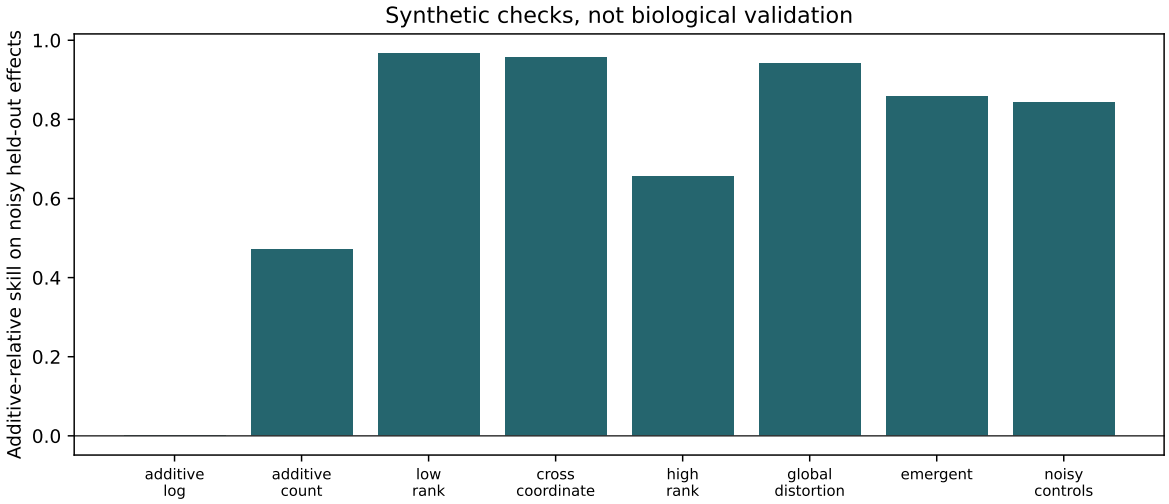


Figure 2: Additive skill across synthetic mechanisms. Positive skill measures predictive improvement, not mechanistic evidence. The shared-distortion case requires comparison to the global correction.

The count-additive generator produces 47.10% skill on the log endpoint, although its underlying count-scale generator has no added interaction. This illustrates scale dependence. In the global-distortion scenario, selected neural skill is 94.13%, but its MSE 0.00110447 exceeds the simpler correction's 0.00104015. A large gain over additivity can therefore coexist with failure of the pair-model claim. The higher-rank scenario selects an unrestricted-output bilinear candidate and yields 65.71% skill; this result does not support a universal low-rank restriction. Emergent and

noisy-control scenarios produce 85.95% and 84.24% skill within their constructed universes.

Across the five random splits of the low-rank generator, selected skill ranges from 96.58% to 97.42%. Five partner folds yield 92.97%–96.63% skill, with 86 or 105 test pairs per fold. Generalization survives this harder synthetic split, but variation across held target groups and overlap among test sets limit interpretation. These results cannot substitute for real-data partner holdouts.

4.2 Norman improves over additivity without a clear structural advantage

Dataset/scale	Neural family	Additive MSE	Neural MSE	Simpler MSE	Skill
Norman (log)	network	0.00852	0.00717	0.00727	0.158
Wessels (log)	sum network	0.01394	0.00542	0.00421	0.611
Norman (count)	sum network	1.24605	1.03911	1.25944	0.166
Wessels (count)	sum network	0.26123	0.19684	0.16913	0.246
Chemical (exploratory)	bilinear	0.01381	0.01223	0.00994	0.114

Table 2: Completed primary biological results and available count sensitivity. Architecture and comparator selection use development data only. Skill is a fraction, and errors across expression scales are not directly comparable.

On 38 withheld Norman pairs, the selected rank-two symmetric network has MSE 0.007169187, compared with 0.008515040 for additivity, giving 15.81% skill. Its separately development-selected reduced-rank regression comparator has MSE 0.007268184. The selected network's paired improvement is only 0.0000989974, with descriptive pair-bootstrap interval $[-0.000239746, 0.000393522]$ and target-product-weight interval $[-0.000477775, 0.000691408]$. Both cross zero, so the structural comparison is inconclusive under these conditional summaries.

An ordinary ridge candidate has the lowest observed classical test MSE, 0.006891604, and 19.07% additive skill. This candidate was not the frozen development-selected simpler comparator. Its favorable test score is a descriptive comparison among prespecified evaluated candidates, not grounds to select a replacement after viewing the test. It nevertheless prevents a broad claim that the selected network outperforms the evaluated classical alternatives.

4.3 Wessels improvement is explained more effectively by a global correction

On 18 withheld Wessels pairs, the selected rank-eight network without multiplicative features has MSE 0.005423558, compared with 0.013936912 for additivity. Its additive skill is 61.08%. The separately development-selected global correction has MSE 0.004214949 and skill 69.76%, so the selected network is worse by 0.001208610 MSE. Pair-specific multiplicative structure has not demonstrated an advantage in this primary experiment.

For the network-minus-global comparison expressed as improvement, the descriptive pair-bootstrap interval is $[-0.00155317, -0.00091829]$, and the target-product-weight interval is $[-0.00189104, -0.00070451]$. Both favor the global correction conditional on these estimated profiles. They are not confirmatory population intervals. The additive comparison is positive under both descriptive schemes, but it cannot distinguish a general correction from pair-specific interaction learning. The selected model's omission of multiplicative features further limits the original architectural hypothesis.

The Wessels count-scale sensitivity also selects the network without multiplicative features. Its MSE is 0.19684419, versus 0.26122700 for additivity and 0.16912624 for the development-selected

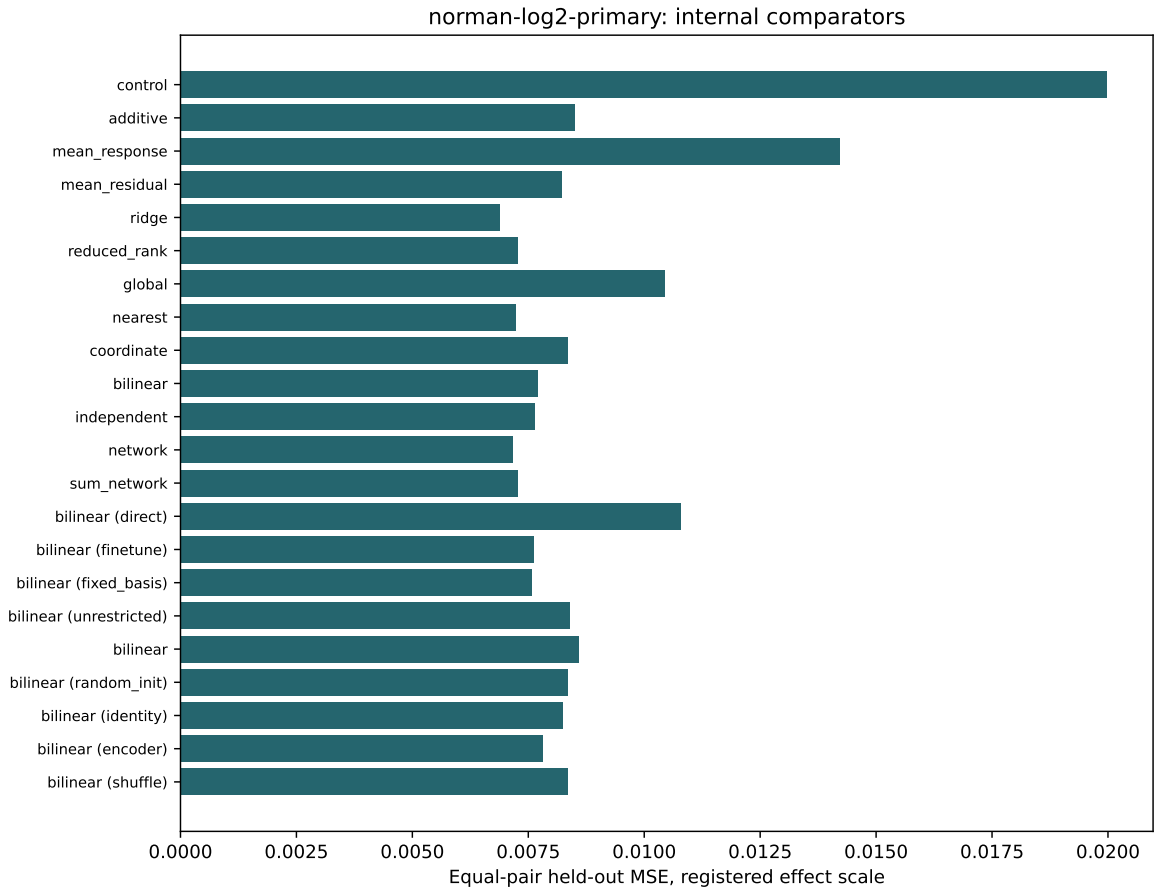


Figure 3: Norman primary internal comparisons. The development-selected neural model and simpler comparator are fixed before test access; the best observed test candidate does not become a new selection.

simpler comparator. Skills are 24.65% and 35.26%, respectively. The negative neural-versus-simpler conclusion persists on this alternative scale, although the magnitudes of count and log errors are not comparable.

Norman count-scale sensitivity selects a rank-eight network without multiplicative features, with MSE 1.03911249 versus additive 1.24604887, giving 16.61% skill. Its development-selected ridge comparator has MSE 1.25943739. The neural improvement over ridge is 0.22032490, but the descriptive pair interval $[-0.06229010, 0.53790781]$ and target-product-weight interval $[-0.26011966, 0.85000429]$ both cross zero. The count-scale target-weight interval for improvement over additivity also crosses zero. Scale sensitivity therefore supports neither a stable product-feature selection nor a resolved classical-versus-neural advantage.

4.4 Reliability and computational diagnostics

Mean split-half residual cosine is 0.2891 for Norman and 0.0838 for Wessels. These technical partitions indicate substantial instability in the estimated interaction residuals; they are neither independent biological replicates nor an upper bound on prediction accuracy. An evaluator-only, swapped-half emergent panel of 20 genes per pair gives MSE 0.2401 and 0.1268, respectively. This

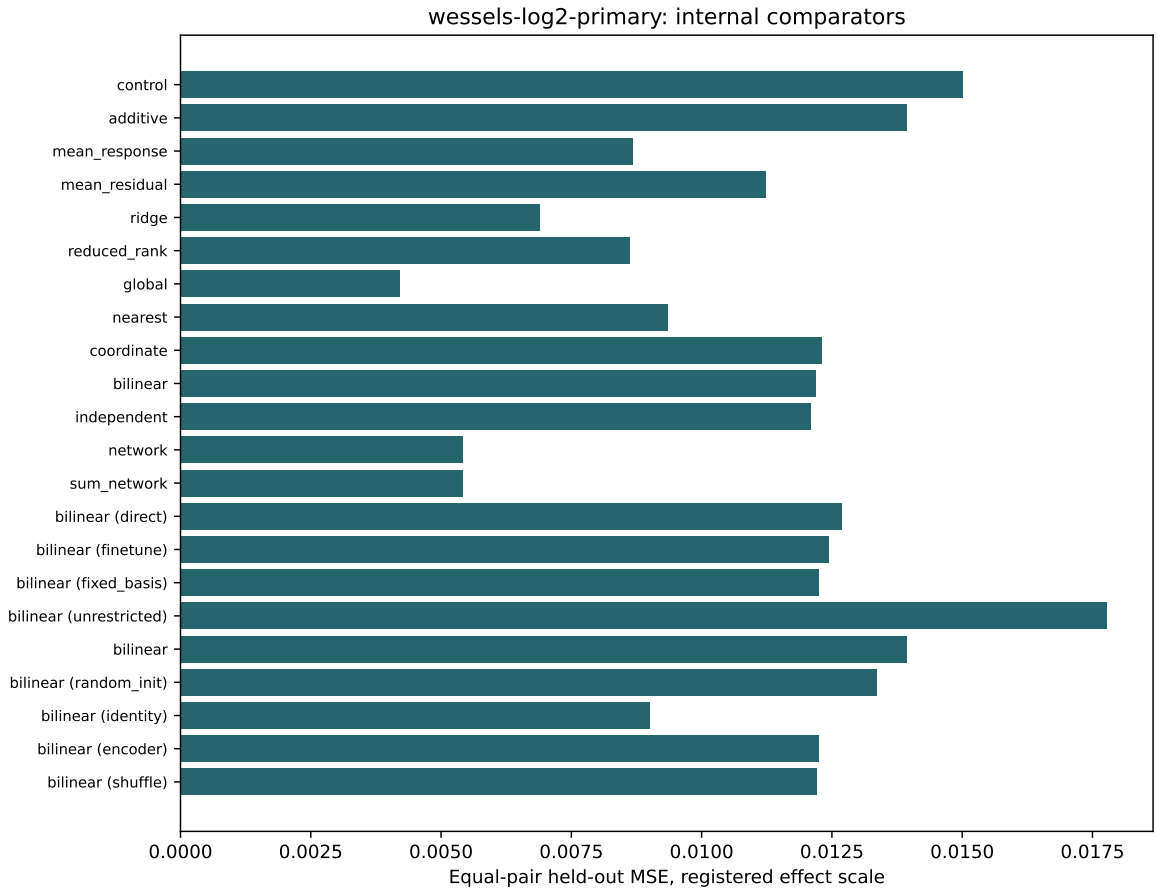


Figure 4: Wessels primary internal comparisons. The global correction outperforms the selected network under the same observed-single input contract.

diagnostic covers only the primary frozen gene universe, so it cannot establish recovery of genes excluded during reference-based filtering.

Ensemble-spread versus pair-error Pearson correlations are 0.5262 and 0.5835 on development out-of-fold predictions for Norman and Wessels, and 0.5441 and 0.3124 on their tests. These associations do not calibrate a failure probability, and no risk threshold is selected. Retrieval against the run's held-out responses identifies 35 of 38 Norman pairs and 17 of 18 Wessels pairs, compared with chance rates 1/38 and 1/18. Retrieval is sensitive to candidate composition and signal strength, so it supplements expression-error evaluation rather than replacing it. Prespecified best, median, and worst error examples are retained without selecting appealing biological narratives.

Selected five-member ensemble checkpoints occupy 6,079,406 bytes for Norman and 5,439,360 bytes for Wessels. With one CPU thread and 100 warm measurements, median single-pair prediction takes 0.676 and 0.751 milliseconds; 95th percentiles are 0.797 and 1.015 milliseconds. Cold checkpoint load takes 47.0 and 45.7 milliseconds. These retained measurements are saved separately as immutable timing records. They describe this machine and artifact layout, not an end-to-end request service. A separate fresh-process measurement after loading and 100 predictions reports peak resident working sets of 222,375,936 and 223,993,856 bytes, including Python and native-library overhead. Training peak memory was not measured. Baseline, capacity, fixed-hyperparameter learning-curve, residual-geometry, reliability, and example figures are generated alongside the re-

sult artifacts.

4.5 Repeated random and combination-partner holdouts

The biological secondary suite completes four additional random-pair seeds per dataset, giving five random splits including the primary, plus five combination-partner holdouts per dataset. Every split independently repeats development selection and final refitting under the same bounded protocol. Partner holdouts exclude development doubles incident to their held targets but retain those targets' measured singles; inner validation uses analogous partner splits. Gene eligibility and normalization remain fixed from allowed references. Overlapping test pairs and shared profiles make these sensitivity runs correlated, rather than independent experimental replications.

Norman neural additive skill ranges from 0.0820 to 0.1581 across random splits and from 0.0111 to 0.1607 across partner splits. Improvement over the independently development-selected simpler method is positive in one of five random splits and four of five partner splits, with partner differences spanning -0.0000812 to 0.0017721 MSE. This protocol dependence does not establish a general structural advantage. Wessels additive skill ranges from 0.5067 to 0.6161 across random splits and 0.2106 to 0.3220 across partner splits; its simpler comparator outperforms the neural selection in all ten runs. These counts are descriptive, not binomial evidence from independent trials.

Dataset	Protocol	Runs	Neural skill range	Gain over simpler range
Norman	random	5	$[0.082, 0.158]$	$[-0.00065, 0.00010]$
Norman	partner	5	$[0.011, 0.161]$	$[-0.00008, 0.00177]$
Wessels	random	5	$[0.507, 0.616]$	$[-0.00162, -0.00097]$
Wessels	partner	5	$[0.211, 0.322]$	$[-0.00139, -0.00025]$

Table 3: Completed correlated biological split sensitivities. Gain is simpler MSE minus selected neural MSE; ranges are extrema, not confidence intervals.

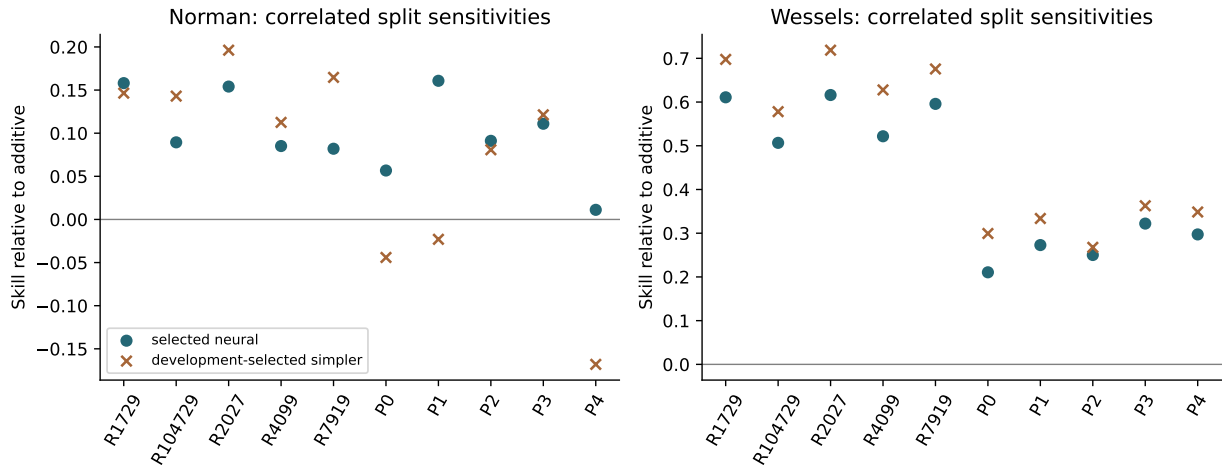


Figure 5: Neural and simpler additive skills across registered biological splits. R denotes random-pair seeds and P partner folds.

4.6 Supplementary adapted public reference

We execute two explicitly adapted SLIM references from pinned upstream revision `5a7e9ade5d0a` (Hu et al., 2026); the full identity and verified source hashes are retained in the method manifest. Its native combination fit is applied to one signed-effect pseudoprofile per condition, retaining the shared gene basis, ridge solve, mean component embedding, and separate single/double biases. We omit the native nonnegative-expression prediction clip because negative effects are valid endpoints. These changes mean the results are adapted references, not reproductions of the published expression protocol. One variant uses 64 supplied STRING dimensions and discloses external information; another substitutes PCA of the allowed observed singles, capped at 16 dimensions.

For each input variant, five development folds select from basis ranks 2, 8, and 16 and ridge penalties 0.1, 1, 10, and 100. Basis fitting uses only singles and that fold's development doubles. These supplementary comparisons were added after the primary results and do not replace the original Opal selection. On Norman, STRING and observed-single variants have MSE 0.00898725 and 0.00782653, respectively, both above the selected Opal MSE. On Wessels their MSE values are 0.00417759 and 0.00418530, both below the selected Opal value and close to the stronger internal global correction.

Dataset	Adapted SLIM input	MSE	Additive skill
Norman	STRING64	0.008987	-0.055
Norman	observed single PCA	0.007827	0.081
Wessels	STRING64	0.004178	0.700
Wessels	observed single PCA	0.004185	0.700

Table 4: Late supplementary signed-effect SLIM adaptations. External STRING information and common-information observed-single PCA are reported separately.

Native GEARS and CPA imports fail in the recorded environment because their required dependencies are absent; CPA's pinned package also declares Python below 3.11, while this research environment uses 3.14. PertAdapt's author endpoint initially returned HTTP 401. The subsequently supplied checkpoint, GO graph and dictionary pass the retained resource audit, but the contextual foundation backbone and native GPU adapter have not been executed. These observations document execution scope, not scientific incompatibility or inferior performance. No native scores are imputed for these methods.

4.7 Extended universe and metric controls

An evaluator-only sensitivity uses all 33,694 original Norman genes and 20,639 Wessels genes. The learned prediction is retained on its frozen schema; outside it, the extension uses observed-single additivity, with no fitted correction. A panel selects the 20 largest absolute residual genes on one technical half and scores the opposite half, then swaps the halves. Outside-schema genes constitute 0.197% of these Norman panel positions and none of the Wessels positions. Selected-extension MSE is 0.240990 and 0.126811, compared with additive values 0.385201 and 0.329619. This sensitivity does not show that a trained extended-universe model would recover emergent responses.

Metric controls include a zero effect, observed additivity, development mean residual, an oracle response as an explicit positive control, and 100 pair permutations of evaluator residuals. These are diagnostics rather than newly eligible candidates. On Wessels, permuted residuals retain mean response Pearson correlation 0.5638 and residual cosine 0.2516 despite MSE 0.020850, above additive MSE 0.013937. The zero-effect prediction also has positive residual cosine 0.6990 while its additive

skill is negative. Positive alignment alone can therefore be driven by shared reference geometry and cannot establish useful prediction.

4.8 Conditional technical measurement sensitivity

Two supplementary 200-draw bootstraps resample cells within each condition and matched stratum while retaining QC, gene eligibility, normalization, fitted weights, and method predictions. Each draw uses one matched-control sample per stratum and one single sample per target, reused for every affected pair. This preserves reference sharing in the resampled profiles. It does not propagate parameter fitting, selection, biological replication, or uncertainty from predicting with newly estimated singles.

For improvement over the development-selected simpler prediction, Norman's mean difference is 0.00010518 with a technical profile percentile range $[0.00004388, 0.00016553]$. Wessels' mean is -0.00120615 with range $[-0.00129652, -0.00113008]$. These ranges describe technical response-estimation sensitivity conditional on fixed predictions. Norman's pair-network uncertainty remains unresolved despite the positive technical range; the schemes answer different questions and must not be combined into a confirmatory confidence interval.

4.9 Chemical confirmation remains unavailable

Original ComboSciPlex GSE206741 metadata records A549 cells, 24-hour exposure, and 1 micromolar per component, consistent with the primary CPA methods (Lotfollahi et al., 2023). Of 25 pair conditions, only seven have both singles in that experiment, covering six compounds. An explicitly exploratory run uses four development pairs with three-fold cross-validation and three held-out pairs, equal-well profiles, and shared plate controls. Its bounded search contains 22 configurations and omits the genetic ablation batch. Nominal embedding dimension is five.

The exploratory chemical selection is a rank-two bilinear model, with MSE 0.01222957 and 11.44% additive skill. A development-selected mean-residual correction has MSE 0.00994341 and 27.99% skill, so the bilinear selection is worse by 0.00228616. Its additive comparison's descriptive interval crosses zero. Three held-out pairs are insufficient for confirmatory inference, model ranking, or the intended chemical method-transfer claim; these scores document source feasibility and another unfavorable structural comparison. The chemical release gate remains unresolved. A replacement or source-qualified external-single match must be registered before examining new chemical test results; unsupported profiles cannot be reconstructed from held-out pairs.

5 Discussion

The Wessels result is useful because it separates two claims. Observed-single additivity benefits from correction under this protocol, but the selected neural correction does not surpass the stronger global baseline. A shared expression distortion or broad response calibration can account for predictive gains without evidence that projected pair products capture additional biology. The global synthetic control reproduces precisely this distinction. Its role is diagnostic, rather than proof that the same mechanism generates the biological data.

Output rank and representation quality remain empirical choices. The synthetic unrestricted-output selection shows that a low-rank constraint can be inappropriate, while synthetic low-rank recovery establishes only that the pipeline can exploit a compatible generator. The bounded real-data rank search does not characterize the full spectrum of reproducible biological residuals. Similarly, training-time or parameter-count savings require total resource accounting, including embeddings,

single reference profiles, ensemble members, and upstream resources. An inexpensive adapter alone does not establish a small inference system.

The primary endpoint covers the frozen reference-informed universe, but a low overall error can hide failure on emergent genes or poorly measured conditions. Correlation is descriptive and cannot establish accurate residual prediction. Additional endpoints and positive metric controls are required to connect aggregate error to experimental usefulness (Miller et al., 2026). Neither selected correction coefficients nor output directions are validated biological mechanisms.

6 Exploratory development amendment

After inspecting the completed primary and secondary findings, we investigated a closed-form symmetric residual kernel procedure. The candidate set contains a shared offset, reference-scaled linear global correction, RBF summed-single and symmetric pair-product kernels, and mixtures of global and pair-product corrections. Inner development folds select among 67 configurations. This procedure permits a shared correction to win; the original primary selection restricted the selected method to a neural family. Historical model selections and test results remain unchanged.

On each genetic development set, three seeds define five outer random folds and five outer partner folds, with three inner folds of the corresponding protocol per outer fit. An expanded linear comparator selects among 25 configurations comprising a shared offset and global, ridge, and rank-four reduced-rank models at eight penalties. The kernel procedure reduces outer-validation MSE relative to this comparator by 1.78–2.70% for Norman random holdouts, 1.13–3.38% for Norman partner holdouts, 2.50–3.98% for Wessels random holdouts, and 0.83–1.00% for Wessels partner holdouts. These are development-only point estimates across dependent resamplings, not independent confirmation. The search spaces and budgets differ, so these results do not establish a capacity-matched structural advantage.

Partner folds overlap; the primary nested score weights validation occurrences equally. Averaging repeated predictions changes the estimand and can reverse Wessels' comparison against the reference-scaled shared global control. No universal advantage over linear methods is inferred. The historical test findings motivated this amendment, and no new historical evaluator responses were loaded. Safe candidate checkpoints reproduce the saved unscored query predictions. Native frontier comparisons, independent confirmation, and novelty remain unresolved. Evidence is recorded separately under `results/improvements/`; it does not replace the frozen primary experiments.

A separately recorded exploratory check adds 45 frozen scFoundation positional-gene-table kernel candidates to the measured-single search, with an identically searched shuffled-gene-table control. These vectors are checkpoint gene positions, not contextual backbone outputs or a native PertAdapt replication. Norman retains the same measured-single candidates in every outer fold. On Wessels, the real table's small random-holdout improvement is matched or exceeded by shuffling gene identities, and its partner score is slightly worse. This does not support an identity-specific benefit; the measured-single candidate remains the exported model. Pretraining-data overlap is unknown.

7 Architecture and coverage amendment

All studies in this section were designed after the historical outcomes were known and use development responses only. They are exploratory. Nested five-outer/three-inner random and partner folds select configurations, while historical evaluator responses remain inaccessible. The primary

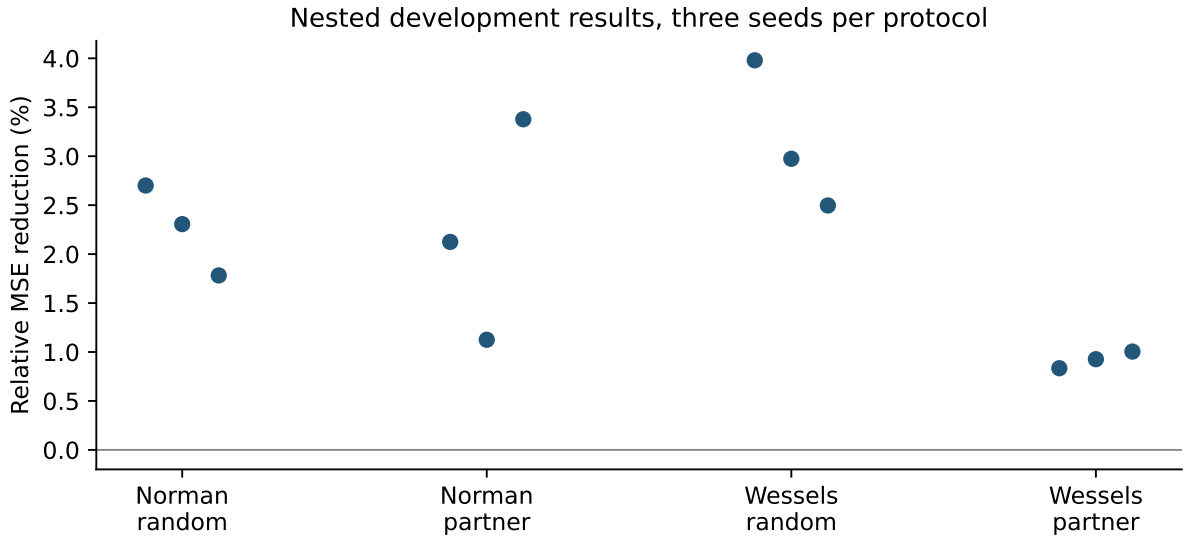


Figure 6: Exploratory nested-development improvement relative to a separately tuned expanded linear search. Each point is one split seed; the points are dependent and are not confidence intervals.

nested score weights validation occurrences equally; overlapping partner folds and repeated seeds are dependent.

7.1 Reference uncertainty, priors and structured outputs

Six implementation priorities are tested: technical reference uncertainty, multiple biological priors, structured output directions, a bounded signal-sensitive loss, independent single-screen transfer, and strong classical experts. Sixteen joint technical bootstrap draws reuse the same matched-control draw within each stratum and the same single-target draw across affected pairs. QC, gene eligibility and normalization remain fixed. Reference reliability shrinks measured-single features and supplies jointly sampled training augmentation. It does not estimate biological variance.

The source bank combines measured-single PCA, a fixed random projection of the supplied GO graph, Hallmark pathway membership (Liberzon et al., 2015), and, in the external-prior tracks, frozen GenePT and ESM2 embeddings from the verified PRESAGE author archive (Genentech, 2025). Missing sources are masked. Neural heads fuse sources and form symmetric summed, absolute-difference and product features. Output directions comprise training-fold residual PCA, pathway-local single-reference directions, regularized gene-local GO directions, and optional external directions. The gene-space loss is computed exactly through the output-basis quadratic. Its bounded signal weights incorporate training response magnitude and technical variance; this is an Opal objective, not an implementation of another study's published weighted metric.

Twelve neural ablations, each with two mixture choices, are compared with the 67-configuration residual kernel and a separately tuned prior-feature ridge. Ablations include uncertainty, fusion, pooling, structured modules, shared output rank and signal weighting. The kernel is selected in all forty competitive outer folds across the original and expanded datasets. This selection permits a simple expert to win. It does not establish a matched-search-budget architecture advantage. The separately nested measurement GP and training-only residual retrieval are worse than the kernel on the original datasets. GP conditional latent intervals omit query measurement variance, full reference covariance and selection uncertainty, so their empirical coverage is diagnostic only.

A frozen TabICLv2 regressor (Qu et al., 2026) predicts eight training-fold residual-PC scores

Dataset	Fold	Kernel	Neural selection	GP	TabICLv2
Norman	random	0.006292	0.006809	0.007012	0.007443
Norman	partner	0.007181	0.007538	0.007878	0.008529
Wessels original	random	0.005032	0.006990	0.008983	0.007458
Wessels original	partner	0.009868	0.011249	0.012159	0.012613
Wessels expanded	random	0.006471	0.010523	0.011666	0.012228
Wessels expanded	partner	0.014467	0.016805	0.017884	0.018610
Wessels protected	random	0.007360	0.011987	–	–
Wessels protected	partner	0.016063	0.018735	–	–

Table 5: Development-only architecture comparisons. Every entry uses the same dataset-specific validation population and weights validation occurrences equally. The neural entry is inner-selected. TabICLv2 uses a fixed CPU configuration, not a native perturbation benchmark. Missing entries indicate that a comparator was not run on that track.

from the same symmetric reference/prior features, using four estimators and a pinned, checksum-verified public checkpoint. Its error exceeds the kernel on all six original/broad-expanded dataset-protocol comparisons. Expanded partner folds emitted Yeo–Johnson overflow warnings, although retained predictions were finite. This fixed configuration does not characterize the best tuned foundation-model performance, and pretraining contamination cannot be excluded.

7.2 Measurement-aware kernels and identity controls

Heteroscedastic kernels fit a generalized-least-squares shared correction with an unpenalized intercept. Each training pair's technical residual variance determines a bounded diagonal noise factor; no validation variance enters the fitting likelihood. Inner folds choose noise strengths zero, one-half or one over the 67 kernel candidates. The unweighted search has 67 candidates, while real-variance and permuted-variance searches each have 201. The matched placebo permutes variance values within training folds only. Shared bootstrap draws preserve dependence when estimating variances, but the fitted likelihood uses their diagonal approximation.

Dataset	Fold	Real noise gain (%)	Permuted gain (%)
Norman	random	1.27 to 2.54	-0.06 to 0.19
Norman	partner	-0.58 to 1.13	0.00 to 0.00
Wessels original	random	0.21 to 1.05	-0.08 to 0.10
Wessels original	partner	-0.07 to 0.19	-0.01 to 0.00
Wessels protected	random	0.92 to 1.24	-0.05 to 0.02
Wessels protected	partner	-0.40 to 0.05	-0.01 to -0.00

Table 6: MSE reduction versus the unweighted kernel over three dependent split seeds. Ranges are extrema, not confidence intervals. Real-variance and permuted controls have the same search budget.

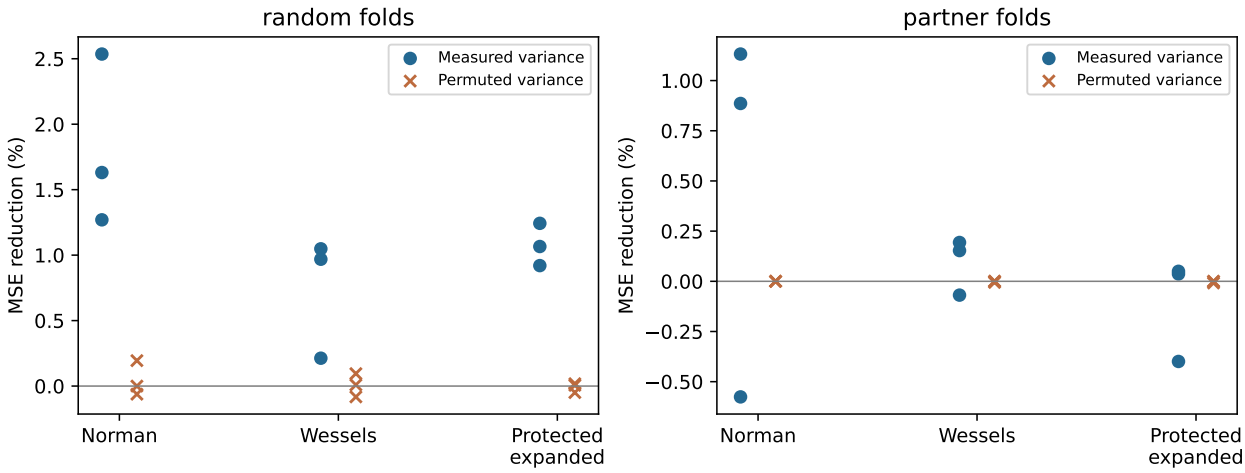


Figure 7: Each point represents a development split seed. Measurement weighting consistently helps random folds in these runs; partner gains are mixed and smaller.

GenePT, ESM2, GO and Hallmark kernels are also tested against an identically searched shuffled-gene-identity bank. Norman random folds and both expanded random tracks retain the measured-only baseline. Original Wessels random-fold MSE improves from 0.00503184 to 0.00500368 with real priors, but shuffling also improves it to 0.00501316. Norman partner error worsens slightly with real priors. Small improvements on the expanded partner tracks do not establish a general identity-specific benefit.

7.3 Independent representation transfer

The verified PRESAGE archive contains semantic embeddings, splits and ranked DEG lists rather than the required expression matrices. We therefore separately download the scPerturb release of the Replogle K562 essential-gene CRISPRi screen (Replogle et al., 2022) and verify its publisher MD5. It contains 310,385 cells, 8,563 output genes and 2,057 single-target conditions. Of these, 2,002 targets map to the supplied GO dictionary. A seed-fixed 1,601/401 training/validation target split trains a 16-to-32-to-16 encoder for 200 updates against training-target response PCs. Its normalized validation-PC MSE is 0.87670, versus 0.88021 for an alpha-ten ridge and 0.90792 for the training mean. These are source-task development scores.

External effects are mean released-X differences against controls. This scale is not equated with Opal's count-derived log2 pseudobulk scale. Only encoder initialization and aligned response directions transfer to the separately fitted current-context pair decoder; external combination labels are not used. Encoder-only, decoder-only and combined transfer are compared with a matched no-transfer head and linear ordinary/weighted objectives. The first transfer run was interrupted after unscaled semantic feature products destabilized its linear controls. That evidence is retained. A second run retained poor fixed-update linear optimization and was also interrupted. Both records remain available. The final linear controls standardize features using training pairs only, preserve scaler buffers in safe checkpoints, and solve the convex gene-space quadratic by converged conjugate gradient. Their frozen-reference fitting omits stochastic reference augmentation; the ordinary and weighted linear objectives otherwise share features and regularization. Results from the corrected track are generated below.

Encoder initialization modestly improves the matched Norman random-fold neural head, but

its partner score is worse. None of the encoder, decoder or combined-transfer heads beats the kernel on any dataset-protocol aggregate. The weighted linear objective has higher primary MSE than its ordinary counterpart on all eight aggregates. These results limit the transfer and signal-objective hypotheses under the tested representation and regularization.

Dataset	Fold	No transfer	Encoder	Decoder	Both	Linear MSE	Linear weighted
Norman	random	0.008892	0.008549	0.008895	0.008613	0.008281	0.008517
Norman	partner	0.010484	0.010785	0.011279	0.010869	0.009534	0.009743
Wessels expanded	random	0.010658	0.011229	0.010621	0.011237	0.023141	0.028550
Wessels expanded	partner	0.019227	0.022255	0.019632	0.023492	0.415128	0.479667
Wessels protected	random	0.012031	0.012413	0.012093	0.012512	0.027270	0.031739
Wessels protected	partner	0.021088	0.022368	0.021576	0.023081	2.346815	2.903400
Wessels original	random	0.007193	0.007259	0.007775	0.007398	0.006618	0.006678
Wessels original	partner	0.016089	0.015757	0.016792	0.016447	0.010943	0.011011

Table 7: Corrected transfer and convex objective controls on development folds. Linear heads solve a regularized gene-space quadratic on frozen standardized features; neural heads use joint reference augmentation. These are different fitting procedures, and none uses historical query responses.

The unbounded convex controls have particularly poor error on some expanded partner folds. Reference-only diagnostics on the first protected partner fold show a maximum standardized query-feature magnitude of 928.19, compared with 8.31 in training. This motivates a separate late exploratory control that clips standardized features to magnitude eight in both fitting and inference, retaining the same converged ordinary and weighted quadratics. It is designed after the unfavorable observations, and its results do not replace them.

Dataset	Fold	Kernel	Linear MSE	Bounded MSE	Bounded weighted
Norman	random	0.006292	0.008281	0.008190	0.008413
Norman	partner	0.007181	0.009534	0.008810	0.008947
Wessels expanded	random	0.006471	0.023141	0.010730	0.011126
Wessels expanded	partner	0.014467	0.415128	0.017244	0.017678
Wessels protected	random	0.007360	0.027270	0.012413	0.012865
Wessels protected	partner	0.016063	2.346815	0.019325	0.019807
Wessels original	random	0.005032	0.006618	0.006615	0.006675
Wessels original	partner	0.009868	0.010943	0.010921	0.010985

Table 8: Late exploratory bounded-feature controls after observing expanded-fold extrapolation. The standardized feature map clips at magnitude eight during training and prediction; the regularized gene-space objectives are solved to convergence. This is a separate amendment, not a replacement of the unfavorable unbounded controls.

7.4 Expanded coverage under a common validation population

Lowering Wessels reference eligibility to fifteen cells yields 28 singles and 158 matched pairs, compared with the original 16 singles and 58 pairs, while retaining the 10,590-gene schema. A broad exploratory repartition assigns 110 pairs to development and 48 to query, moving twelve historical query identities into development. A second expansion preserves all original development/query identity roles and assigns only new pairs. Its development set contains no original query identity. Nevertheless, thirty of its query identities had already appeared in the broad development track; it is not independent confirmation.

A controlled comparison scores the same forty original development pairs under the same outer folds and gene panel. Additional protected development pairs may enter training, but extra pairs containing excluded inner or outer partner targets are removed. Extra singles alone change random error by only 0.03% and worsen partner error by 0.38%. Extra eligible pair labels reduce random error by 16.05% and partner error by 4.92%. Adding measurement weighting gives 18.73% and 4.91% reductions. This supports a coverage benefit for this exploratory population, not a new architecture mechanism or independent biological result. Across three dependent split seeds, extra pair labels reduce random-fold error by 16.05–17.60% and partner error by 4.00–4.99%. With measurement weighting the ranges are 18.73–19.54% and 4.08–4.99%, respectively. These ranges are split sensitivities, not confidence intervals. A stricter sensitivity removes extra pairs touching any validation component and gives smaller partner gains; both policies are retained.

Fold	Original	Extra singles	Extra pairs	Pairs + noise
random	0.005032	0.005030	0.004224	0.004089
partner	0.009868	0.009905	0.009382	0.009383

Table 9: Common-population Wessels nested-development MSE. Original query identities are excluded from all training. Expanded response histories are known, so these scores are exploratory.

The common `main.py frontier` entry point prepares resources, repeats registered studies, validates exported inference, generates evidence tables and builds this multi-file manuscript. Large licensed inputs, prediction banks and derived weights remain local and excluded from source-control commits. Safe model exports include reference banks, numerical states, immutable gene order, checksums and contextual input guards. These implementation and empirical gains do not satisfy an independent frontier-quality claim.

8 Limitations and completion criteria

This manuscript reports an internal benchmark, not a completed Opal v1 release or a submission-ready validation study. A sufficiently sized independent chemical benchmark, native public-method runs, expanded capacity matching, fitting-aware and replicate-aware uncertainty, and biological interpretation remain necessary. Repeated biological splits, count-scale sensitivities, adapted SLIM references, extended-universe diagnostics, technical-cell bootstraps, and a seven-pair chemical feasibility experiment are complete, but do not resolve these gates. The completed reliability diagnostics do not establish a usable abstention policy. The small Wessels held-out set and restricted single coverage particularly limit generalization.

Randomness from optimization is represented by finalist and ensemble seeds, but the reported bootstrap sensitivities do not include the full selection or refitting process. The technical-cell bootstrap holds predictions fixed, while pair and target-product bootstraps hold estimated profiles fixed. Lane and gemgroup matching reduce a source of mismatch without creating biological replication. Cell counts improve estimates of condition means; they do not increase the number of independent perturbation experiments. Source licensing also requires dataset-specific review before redistribution.

The implemented protocol rejects retrospective replacement of an unfavorable selected model. Further tuning must create a new documented run and preserve the current result. The full research claim will depend on whether improvements survive stronger same-information alternatives and independent contexts. Current evidence supports synthetic recovery, an inconclusive Norman

structural comparison, and a negative Wessels comparison despite gains over additivity in both genetic systems.

Data and code availability

The local research repository contains the common `main.py` entry point, source adapters, registered configurations, tests, source manifests, development selections, held-out predictions, and result-generation code. Dataset source identities and checksums are recorded with prepared artifacts. This draft does not assert an archived public code release, a public checkpoint URL, or a data redistribution license. Norman is sourced from Zenodo record 7278143 v1.1; Wessels and chemical metadata are sourced from GEO GSE213957 and GSE206741. Hugging Face hosting and demonstration applications are outside this research repository's scope. A public archive identifier and finalized reuse statement remain to be supplied before submission.

Ethics declaration

The reported work reuses published cell-line perturbation data and conducts computational experiments. It reports no new human recruitment, animal experiments, or prospective intervention study. Original experimental oversight is described in the source publications. No new ethics approval or exemption is claimed here. The predictions have not been validated for clinical or therapeutic use.

Author contributions, funding, and competing interests

Author names, affiliations, CRediT contributions, funding sources, and competing-interest declarations have not been provided. These declarations remain incomplete; no absence of funding or competing interests is inferred. They must be completed by the responsible authors before submission.

AI assistance disclosure

OpenAI Codex assisted with repository implementation, experimental orchestration, and manuscript drafting. Numerical claims in this draft refer to recorded experiment artifacts rather than generated estimates. AI assistance does not establish authorship, independent replication, or expert validation. Responsible authors must verify the code, analysis, source attribution, and final text, and adapt this disclosure to the selected venue's policy.

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A Primary diagnostic figures

The following plots are generated from frozen primary artifacts. Capacity compares registered candidates rather than an exhaustive capacity search. Learning curves keep full-development-selected hyperparameters fixed; split-half spectra describe technical measurement reproducibility, not biological replication or calibrated risk.

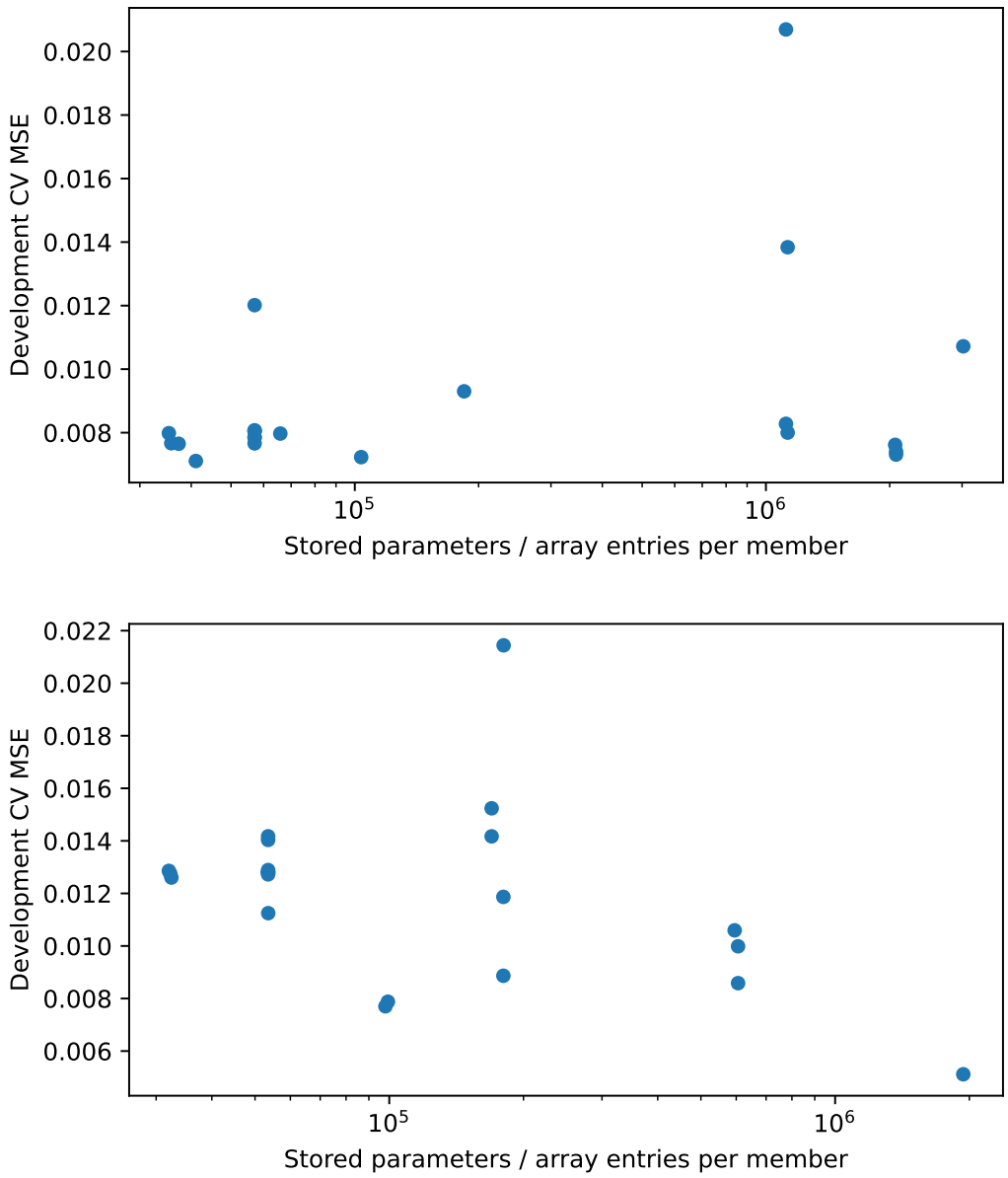


Figure 8: Registered capacity comparisons for Norman (top) and Wessels (bottom). Parameter counts do not include complete reference-bank and library memory.

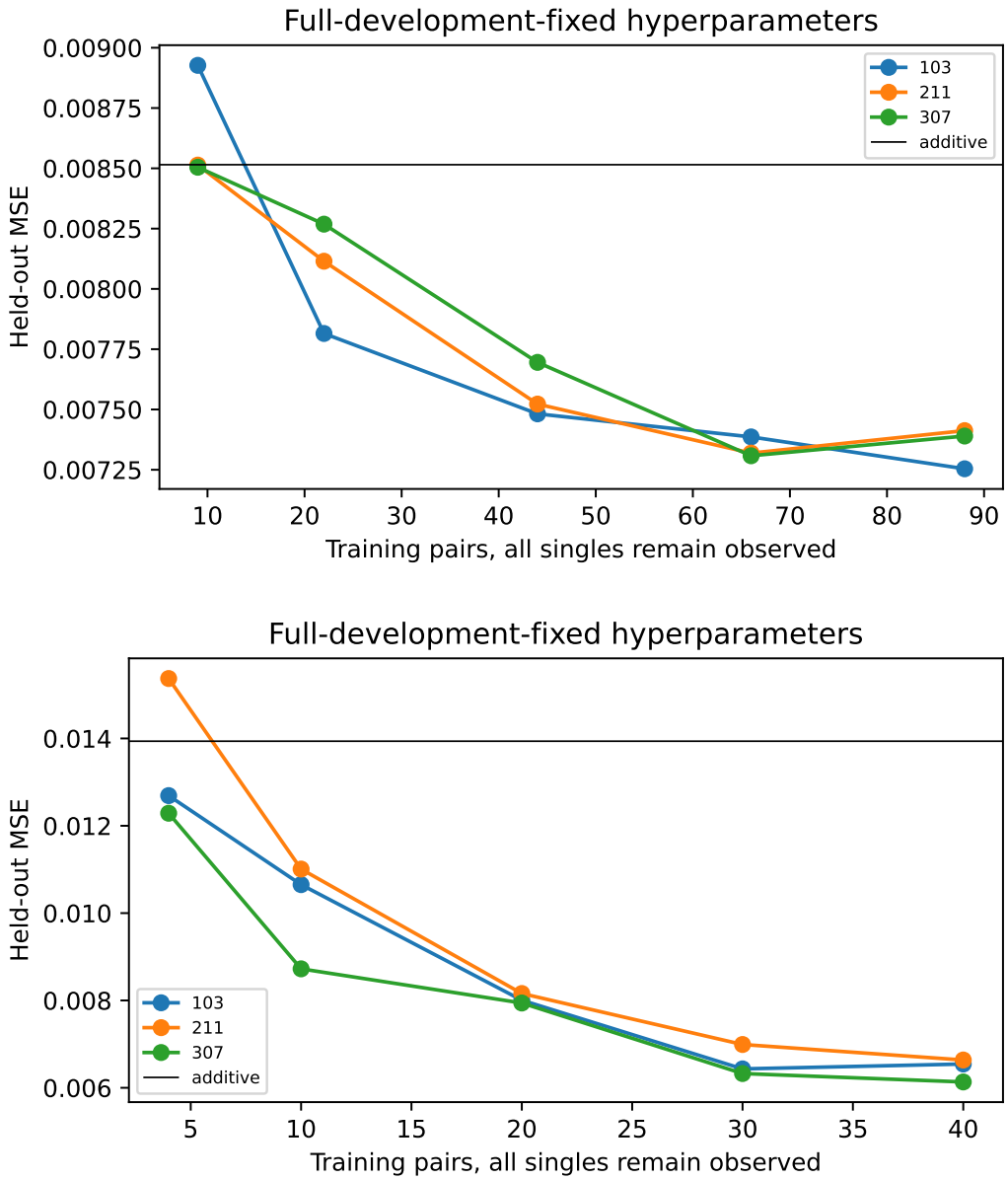


Figure 9: Fixed-hyperparameter development-subset sensitivity for Norman (top) and Wessels (bottom). Three nested subset orders share the same held-out evaluation and are correlated.

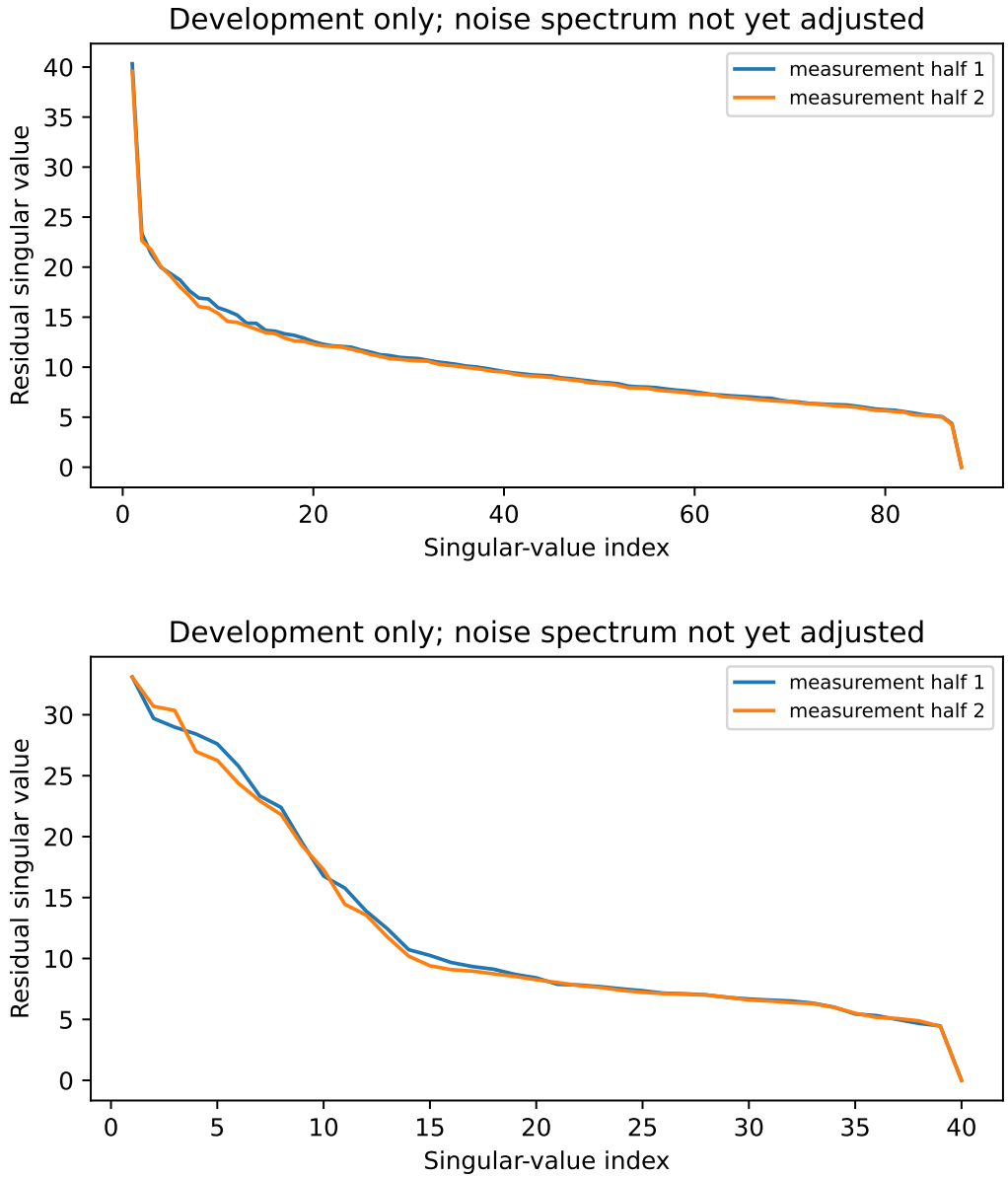


Figure 10: Development residual spectra from the two technical halves for Norman (top) and Wessels (bottom). Noise has not been subtracted; these do not identify biological interaction mechanisms.

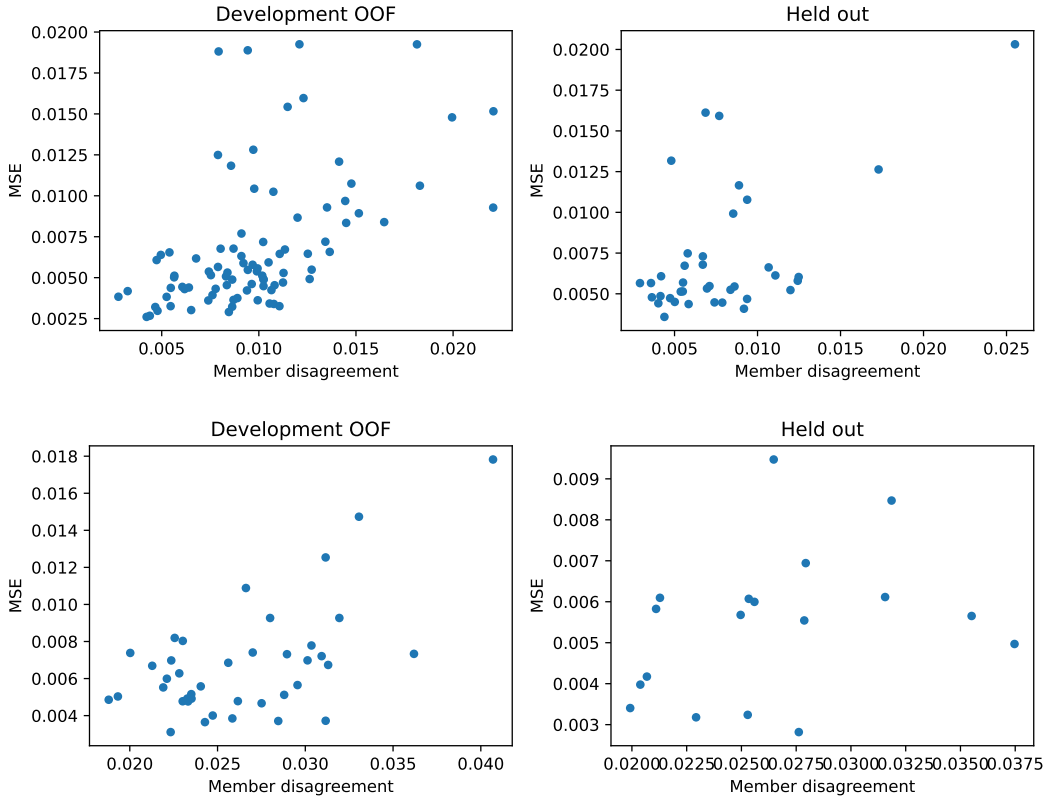


Figure 11: Uncalibrated ensemble-spread versus condition-error diagnostics for Norman (top) and Wessels (bottom), shown on development out-of-fold and held-out predictions. Association does not establish a calibrated failure probability.